Classification of the Mixed Radial–Support Composition Inequality as Printed

Candidate full solution to the second alternative in arXiv:2005.11253, Problem 4.1

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Status. Candidate full solution of the *second alternative as printed.* In dimensions $n \geq 2$, it holds for all K_1, K_2 exactly for centered Euclidean balls $T = sB_2^n$. The printed final term is $T \circ K_2$, not $T \odot K_2$; this distinction is essential.

1 Source question and scope

Milman and Rotem define two compositions of convex bodies containing the origin [1]. If r_K and h_T denote radial and support functions, then

$$T \odot K = \operatorname{conv}\{r_T(u) r_K(u) u : u \in S^{n-1}\},$$
$$T \circ K = \operatorname{conv}\{h_T(u) r_K(u) u : u \in S^{n-1}\}.$$

Problem 4.1 on source PDF page 13 asks, among other things, for which T the following inequality holds for all K_1, K_2:

$$|T \odot (K_1 + K_2)|^{1/n} \geq |T \odot K_1|^{1/n} + |T \circ K_2|^{1/n}. \tag{1}$$

Here and below $|\cdot|$ is n-dimensional volume. The mixture of $\odot$ and $\circ$ on the right is exactly what appears in the source.

Note that if $T = B_2^n$ then both compositions preserve K, i.e. this is the identity map on $\mathcal{K}_0$.

Problem 4.1. *For which bodies T do we have a volume inequality of the form*

$$|T \circ (K_1 + K_2)|^{\frac{1}{n}} \geq |T \circ K_1|^{\frac{1}{n}} + |T \circ K_2|^{\frac{1}{n}}$$

or

$$|T \odot (K_1 + K_2)|^{\frac{1}{n}} \geq |T \odot K_1|^{\frac{1}{n}} + |T \circ K_2|^{\frac{1}{n}}?$$

For example, when $T = B_2^n$ these inequalities are obviously true, as they reduce to the standard Brunn-Minkowski inequality. Hence one may study e.g. the case when T is very close to the Euclidean ball.

Figure 1: Problem 4.1 on source PDF page 13. The last right-hand term in the second display is $T \circ K_2$.

We use the source convention: a convex body is compact and full-dimensional, and it is only required to contain the origin. A separate one-dimensional statement is included.

2 Proof intuition

The asymmetry makes the problem rigid. Insert $K_1 = tB_2^n$ and $K_2 = B_2^n$. Since $T \odot B_2^n = T$, homogeneity forces T to have at least the volume of $T \circ B_2^n$. But $r_T \le h_T$ pointwise, so $T \subseteq T \circ B_2^n$. The nested bodies must therefore be equal. Every radial generator $h_T(u)u$ of the larger body now lies back in T, forcing the reverse inequality $h_T \le r_T$.

Thus $r_T = h_T$. In dimension at least two this common function is constant: subdivide a spherical geodesic into very short arcs and compare successive radii using the support inequality. Hence T is a centered ball. The converse is the ordinary Brunn–Minkowski inequality.

3 Formal result

For fixed T, write

$$R_T(K) = T \odot K, \qquad H_T(K) = T \circ K,$$

and

$$F_R(K) = |R_T(K)|^{1/n}, \qquad F_H(K) = |H_T(K)|^{1/n}.$$

Theorem 1. *Let $n \ge 2$ and let $T \subset \mathbb{R}^n$ be a convex body containing 0. Then (1) holds for every pair of convex bodies K_1, K_2 containing 0 if and only if*

$$T = sB_2^n$$

for some $s > 0$.

For $n = 1$, every nondegenerate interval containing the origin satisfies the printed inequality, always with equality.

Proof of Theorem 1. Assume first that (1) holds for every K_1, K_2. Both compositions are positively homogeneous in their second argument. Substitute $K_1 = tB_2^n$ and $K_2 = B_2^n$, where $t > 0$. Since $tB_2^n + B_2^n = (t+1)B_2^n$, the printed inequality becomes

$$(t+1)F_R(B_2^n) \ge tF_R(B_2^n) + F_H(B_2^n).$$

Therefore

$$|T|^{1/n} = F_R(B_2^n) \ge F_H(B_2^n), \tag{2}$$

where we used $R_T(B_2^n) = T$.

For each $u \in S^{n-1}$, the radial boundary point $r_T(u)u$ belongs to T. Consequently

$$r_T(u) = \langle r_T(u)u, u\rangle \le h_T(u).$$

The defining radial generators therefore give

$$T = R_T(B_2^n) \subseteq H_T(B_2^n). \tag{3}$$

Together with (2), this gives equal volumes. Both sides of (3) are full-dimensional convex bodies, so

$$T = H_T(B_2^n) = \operatorname{conv}\{h_T(u)u : u \in S^{n-1}\}. \tag{4}$$

In particular, every generator $h_T(u)u$ in (4) belongs to T. By maximality of the radial function, $h_T(u) \le r_T(u)$. Combined with the reverse pointwise inequality already proved, this yields

$$r_T(u) = h_T(u) =: \rho(u) \qquad (u \in S^{n-1}). \tag{5}$$

If $\rho(u) = 0$, then applying the next inequality in both directions shows that ρ vanishes in a spherical neighborhood of u. Because $\rho = h_T$ is continuous, its zero set is also closed. The sphere is connected for $n \geq 2$, and T is full-dimensional, so ρ has no zeros.

It remains to show that ρ is constant. Since $\rho(u)u \in T$, the support inequality in direction v gives

$$\rho(u)\langle u, v\rangle \leq h_T(v) = \rho(v). \tag{6}$$

Let $u, v \in S^{n-1}$, and subdivide a shortest spherical geodesic from u to v into N equal arcs of length $\delta < \pi/2$. Applying (6) in both directions to consecutive subdivision points and multiplying yields

$$(\cos\delta)^N \leq \frac{\rho(v)}{\rho(u)} \leq (\cos\delta)^{-N}.$$

Here $\delta = \alpha/N$, where $\alpha \leq \pi$ is the geodesic length. Since $(\cos(\alpha/N))^N \to 1$, letting $N \to \infty$ gives $\rho(v) = \rho(u)$. Thus $\rho \equiv s$ and $T = sB_2^n$.

Conversely, if $T = sB_2^n$, then $r_T = h_T = s$, and therefore

$$R_T(K) = H_T(K) = sK.$$

The desired inequality is exactly the classical Brunn–Minkowski inequality [2], multiplied by s.

Finally let $n = 1$. Write $T = [-a, b]$ and $K = [-c, d]$, with positive endpoint lengths. On the two directions of S^0, radial and support functions agree, and direct substitution gives

$$R_T(K) = H_T(K) = [-ac, bd].$$

Its one-dimensional volume is $ac + bd$, which is additive under Minkowski sums of intervals. Hence the printed inequality is equality for every interval T containing the origin. $\square$

4 Verification and limitations

The proof is non-computational. The deterministic script in `code/numerical_reconnaissance.py` checks 22,324 polygon pairs across support and radial modes and finds failures for several nonball multipliers. It was used only during discovery and is not evidence for the theorem. It may be run from the packet directory with `conda run --no-capture-output -n sandbox python code/numerical_reconnaissance.py`.

The theorem resolves the second alternative exactly as typeset in Problem 4.1. It does *not* address the first, all-$\circ$ inequality. It also does not address the symmetric radial variant obtained by replacing the final $T \circ K_2$ in (1) with $T \odot K_2$. If that replacement was the authors' intent, the present result should be read as a classification of the literal statement and as evidence that the typographical distinction is mathematically consequential.

Run indexes were searched for arXiv:2005.11253, the paper title, composition of convex bodies, and the mixed radial/support formula. A bounded arXiv search on 19 July 2026 found the source but no later paper resolving this exact as-printed alternative. Human review should focus on the source-formula scope and the implication $T = \mathrm{conv}\{h_T(u)u : u \in S^{n-1}\} \Rightarrow r_T = h_T$.

References

[1] V. Milman and L. Rotem, *Novel view on classical convexity theory*, arXiv:2005.11253v1, 2020.

[2] R. J. Gardner, *The Brunn–Minkowski inequality*, Bull. Amer. Math. Soc. (N.S.) **39** (2002), 355–405.